

Vincent Lau

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PROFESSIONAL EXPERTISE

SQL | Python | Spark | GCP | Azure | ETL | Airflow | Power BI | Kafka | Jira | Git | Data Modeling

WORKING EXPERIENCE

Sr. Data Engineer, Analytics

Glocal Foundation, Inc

Vancouver, CA

May 2024 – Present

- Designed **partition keys** to optimize the **ETL** process and built a **data lake** on **GCS bucket** using **Delta Lake** to manage HR data, including demographic, geolocation and PII. Ensured **ACID Transactions**, leveraged highly compressed **Parquet** format, and reduced storage costs by over **\$2,000** annually
- Optimized **Spark** data transformation by utilizing **repartition** and **broadcast join**, enhancing data processing efficiency and reducing dataset landing time by **50%**
- Orchestrated data processing workflow using **GCP Cloud Composer** to trigger data ingestion pipeline from external API to **GCS bucket** and utilized **Spark** to speed up data processing, designed trigger rules and **DAG schedules** with **crontab**
- Developed a centralized interactive dashboard via **Power BI**, integrating **10+ tables** and conducting **deep-dive analysis**, enhancing HR strategy and decision support

Data Engineer, Analytics II

Defend Technology, Inc

Vancouver, CA

2023 – 2024

- Designed and implemented large-scale data pipelines and data warehousing solutions using **GCP Big Query** for data storage and analysis, **PySpark** for data processing, and **Airflow** for orchestration, enabling Defend to gain insights into user behaviour
- Implemented **GCP Dataflow** pipeline to ingest streaming data from **Kafka** into **Big Query** with exactly-once delivery, enabling real-time analysis for personalized matchmaking recommendations and enhancing user engagement and for a Dating App (65 Square)
- Collaborated with cross-functional teams using **Agile** methodologies and **Jira** to identify business requirements and design efficient data models supporting analytics and reporting needs

Data Engineer, Analytics I

CITIC Bank | Rainer Consulting

Beijing, China

2019 – 2023

- Engaged banking **Digital Transformation** and achieved **Target Marketing** project, built data warehousing through developing data mapping document and **ETL Stored Procedures** in **DB2**
- Automate data loading and reporting processes through **Shell** scripts resulting in **1 FTE saving**
- Migrate native **DB2** central data warehouse to **Azure Synapse Analytics**, enabling advanced analytics capabilities for downstream teams and reducing infrastructure cost by **30%**
- Collaborating with data scientists to build **ML** models, including developing a **K-Means** clustering for user segmentation, and using **RFM** and **Linear Regression** model to attract new customers and increase deposit

EDUCATION

University of Nottingham

M.S. in Artificial Intelligence

Nottingham, UK

Sep 2021 – Dec 2022

Guangxi Normal University

B.S. in Computer Science

Guilin, China

Sep 2016 – Jul 2020